

Aryan Kakkar

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EDUCATION

Wilfrid Laurier University

Sep. 2023 – Aug. 2027

Honours Bachelor of Science in Computer Science (Co-op), Minor in Economics

Waterloo, ON

- Concentration in Big Data Systems
- Relevant Coursework: Probability & Statistics (R), Linear Algebra, Algorithms, Data Structures, Machine Learning, Databases, Data Visualization, Canadian Banking & Financial Systems

EXPERIENCE

Machine Learning Engineer Intern

Sep. 2025 – Dec. 2025

Multiverse Computing

Toronto, ON

- Built and maintained **end-to-end production data pipelines** to ingest, preprocess, and aggregate large-scale, multi-source sports performance datasets covering **700+ academy and first-team players using Python**.
- Implemented **simulation and modeling pipelines**, including Monte Carlo-based workflows with Gaussian uncertainty modeling, to evaluate outcomes across longitudinal data in a reproducible and scalable manner.
- Integrated and evaluated multiple modeling approaches, including **quantile regression and XGBoost**, within shared pipelines to enable benchmarking and comparative experimentation.
- Contributed to an internal **LLM-based multimodal edge project**, implementing **speech-to-text (STT)**, **text-to-speech (TTS)**, and **voice activity detection (VAD)** for interactive workflows.
- Developed internal tools and utilities to support experimentation, inspection, and debugging of pipeline outputs, improving **iteration speed and developer productivity**

Director of Logistics

Sep. 2025 - Dec. 2025

Laurier Cricket Club

Waterloo, ON

- Streamlined **logistics workflows** and **cross-department coordination** to improve planning and execution of club operations and matchdays
- Supported **VP Operations** in ensuring smooth execution of events, matchdays, and internal processes

Project Developer

Sep. 2024 - Sep. 2025

Laurier Cybersecurity Club

Waterloo, ON

- Built **analytics features** for a **full-stack platform**, enabling **real-time tracking of event and user metrics** through integrated backend APIs and front-end dashboards.
- Designed and implemented **data flows** between backend services and visualization layers, ensuring accurate, timely reporting of **operational metrics**
- Collaborated in an **Agile team environment**, contributing to sprint planning, debugging, and **peer code reviews** using **GitHub** to improve system reliability and data consistency

Summer Camp Counselor & BACP Supervisor

Jul. – Aug. 2023 & 2024

City of Richmond Hill – Rouge Woods & Richmond Green Community Centers

Richmond Hill, ON

- Coordinated **scheduling and logistics** for programs serving up to **100 participants**, optimizing daily operations
- Adjusted program planning using **participant feedback** and **attendance trends** to improve flow and engagement

Customer Service Representative

Aug. 2021 – Aug. 2022

Farm Boy Inc.

Richmond Hill, ON

- Delivered customer-facing recommendations and supported inventory and store operations through collaboration

PROJECTS

Personal Portfolio

Oct. 2025 - Dec 2025

- Designed and built a **responsive web application** using **React**, **JavaScript**, and **HTML/CSS** to showcase projects, experience, and technical skills with a focus on clarity, usability, and accessibility
- Improved performance and user experience through iterative testing, structured component design, and **version-controlled development** using **Git** and **GitHub**, incorporating accessibility-focused UI improvements

Budget Planner

Apr. 2025 - Jul. 2025

- Built an **interactive financial dashboard** using **React**, **JavaScript**, and **CSS** to analyze spending patterns and simulate savings scenarios based on user-defined inputs
- Implemented **data aggregation logic** in **JavaScript** to compute monthly totals and category level breakdowns, and created clear visual summaries using **Chart.js**, with version control and collaboration managed by **Git** and **GitHub**

Clicker Game

Dec. 2024 - Jan. 2025

- Developed a browser-based application using **JavaScript**, and **HTML/CSS** featuring **real-time state updates** and dynamic difficulty scaling based on user interaction
- Improved engagement and responsiveness through iterative testing and refinement, maintaining clean **version controlled code** using **Git** and **GitHub**

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, JavaScript, HTML, CSS, TypeScript, VBA, PowerShell

Data & Modeling: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Time-Series Analysis, Monte Carlo Simulation

Visualization & APIs: Matplotlib, Streamlit, Chart.js, FastAPI, Flask

Developer Tools: Git/GitHub Actions, JupyterLab, VSCode, JetBrains, Jest, Figma